CO 4 AT-2

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Comparative Analysis of AI Language Models

Questions 32 and 33

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32. Automated Customer-Support Chatbot

Introduction

A company wants to develop an automated customer-support chatbot. The system should understand customer questions, maintain conversational context, generate useful responses, and connect with company information and business services. A suitable architecture must provide accurate language understanding, natural response generation, scalability, reliability, and integration with external knowledge sources.

Comparison of Four Models

Model	Architecture / Main Strength	Suitability for Customer Support
BERT	Bidirectional encoder model mainly designed for understanding text.	Strong for intent classification, FAQ retrieval, sentiment analysis, and information extraction, but not naturally designed to generate complete responses.
GPT	Generative transformer designed for text generation and conversational interaction.	Highly suitable for natural dialogue, contextual responses, instruction following, and integration with external tools and knowledge.
T5	Encoder-decoder model that treats many NLP tasks as text-to-text problems.	Flexible for question answering, summarization, classification, and generation, but a complete chatbot requires additional system

		design.
LLaMA	Open-weight generative language model family supporting customization and local deployment.	Suitable for conversational generation and useful when privacy, self-hosting, or fine-tuning control is important.

BERT

BERT is a bidirectional transformer model designed primarily for understanding language. It can examine context from both directions and is effective for tasks such as intent classification, sentiment analysis, named-entity recognition, question matching, and extracting information from customer messages. In a customer-support system, BERT can help identify what a customer wants and route the request to the correct service.

However, BERT is not primarily a generative model. It is therefore less suitable as the main component for producing long, natural conversational answers. A BERT-based system would normally require additional retrieval, templates, or a separate generative model to construct final responses.

GPT

GPT is a generative transformer architecture designed to predict and generate text. It is highly suitable for conversational applications because it can understand instructions, use conversation context, and produce fluent responses. A GPT-based chatbot can answer common questions, explain procedures, summarize information, and guide customers through multi-step interactions.

GPT can also be connected to external tools and company systems. For example, a chatbot can retrieve order status, create support tickets, search a knowledge base, or provide account-related information. This makes the architecture particularly suitable for practical customer-support automation.

T5

T5 uses an encoder-decoder transformer architecture and frames language tasks as text-to-text problems. The same general architecture can be adapted to question answering, summarization, classification, translation, and text generation. This flexibility makes T5 useful for building specialized NLP components.

For customer support, T5 could be fine-tuned to answer frequently asked questions or transform customer requests into appropriate responses. However, a complete conversational chatbot may require additional components for dialogue management, retrieval, tool integration, and long-term context.

LLaMA

LLaMA is an open-weight generative language model family. It can produce conversational responses and can be adapted to specialized domains through prompting, fine-tuning, or other customization approaches. An important advantage is the possibility of greater control over deployment and data processing.

LLaMA is attractive for organizations that want to host models on their own infrastructure or require greater control over customization. The main challenges are infrastructure requirements, model serving, maintenance, optimization, and ensuring reliable responses in production.

Architecture Comparison

Requirement	BERT	GPT	T5	LLaMA
Natural conversation	Limited	Excellent	Good	Excellent
Language understanding	Excellent	Excellent	Excellent	Excellent
Response generation	Limited	Excellent	Excellent	Excellent
Customization	Good	Good	Excellent	Excellent

Tool / system integration	Requires additional components	Excellent	Requires additional design	Excellent with suitable implementation
Self-hosting control	Good	Depends on deployment	Good	Excellent
Overall chatbot suitability	Moderate as a component	Very high	High with supporting components	High with supporting infrastructure

Recommended Architecture

GPT-based architecture is the most suitable overall solution for an automated customer-support chatbot. A practical architecture can contain a customer interface, conversation manager, GPT model, retrieval component, company knowledge base, and tool integrations. Retrieval can provide current company information, while tools can execute business operations such as order tracking or ticket creation.

- User Interface: receives customer questions and displays chatbot responses.
- Conversation Manager: maintains the relevant conversation context and controls the interaction.
- GPT Model: understands the request and generates a natural response.
- Knowledge Base / Retrieval: provides company-specific information and reduces dependence on model memory.
- Business Tools: connect the chatbot to order systems, ticketing systems, account services, and other applications.
- Monitoring and Safety Layer: checks responses, logs interactions, and supports quality improvement.

Advantages of the Recommended Solution

The GPT-centered architecture provides a strong balance between conversational quality and practical integration. It can respond naturally while also using external information and tools. Retrieval-augmented generation can make answers more specific to company policies and products. The architecture can also be scaled as customer demand increases.

BERT can still be included as a supporting component for intent detection or classification. Similarly, LLaMA may be selected instead of GPT when self-hosting and customization are more important than simplicity of deployment. Therefore, the best architecture depends on organizational constraints, but GPT is the strongest general-purpose choice for an automated support chatbot.

Conclusion

Among the four models, BERT is strongest for language understanding tasks, T5 provides a flexible text-to-text framework, and LLaMA offers strong generation with open deployment and customization possibilities. GPT provides the most suitable overall combination of natural conversation, contextual understanding, response generation, and integration capability. Therefore, a GPT-centered retrieval-augmented customer-support architecture is recommended.

33. Automatic News Summarization

Introduction

A news organization wants to automatically summarize thousands of news articles. Such a system must process a large volume of text, identify important facts, produce concise summaries, and maintain the meaning of the original articles. The selected model should provide strong language understanding and generation while also supporting efficient large-scale deployment.

Comparison of BERT, GPT, T5, and LLaMA

Model	Architecture / Main Strength	Suitability for News Summarization
BERT	Encoder-based bidirectional transformer focused on language understanding.	Useful for extracting important information or ranking sentences, but less suitable for direct abstractive summary generation.
GPT	Generative transformer designed for fluent text generation.	Very strong for flexible, coherent summaries and editorial instructions; suitable when high-quality generation is the priority.
T5	Encoder-decoder sequence-to-sequence transformer.	Highly suitable because summarization is naturally represented as converting an article into a shorter text output.
LLaMA	Open-weight generative language model family.	Strong summarization and high customization/self-hosting flexibility, but

		requires appropriate infrastructure and model management.

BERT

BERT is primarily an encoder-based model designed to understand text. It is effective for classification, information extraction, semantic similarity, and identifying relevant content. In a news summarization workflow, BERT can help determine which sentences or pieces of information are important.

However, BERT is not primarily designed for abstractive text generation. It therefore does not provide the most direct solution when the objective is to generate a fluent summary in new wording. BERT is better used as a supporting component for extraction, ranking, or article classification.

GPT

GPT is a generative transformer capable of producing coherent text from an input prompt. It can summarize news articles, follow instructions about summary length and style, and produce summaries suitable for different audiences. Its strong natural-language generation ability makes it a powerful option for automated summarization.

A GPT-based summarization system can be instructed to preserve important facts, avoid unnecessary details, and produce concise outputs. For thousands of articles, however, the organization must consider processing cost, latency, context length, infrastructure, and quality-control requirements.

T5

T5 uses an encoder-decoder architecture and converts NLP problems into text-to-text tasks. Summarization fits naturally into this framework: the input is a complete article and the output is a shorter summary. This makes T5 particularly appropriate for dedicated summarization systems.

A T5 variant can be fine-tuned using news articles and reference summaries. Fine-tuning can help the model learn the organization's preferred summary length, writing style, and domain vocabulary. Because the task is clearly defined, a specialized T5 model can be integrated into an automated high-volume pipeline.

LLaMA

LLaMA is an open-weight generative language model family that can be used for summarization and other language-generation tasks. It can be adapted to a specific domain and can support local or controlled deployment. This can be useful for news organizations that want greater control over their data and infrastructure.

The main consideration is operational complexity. Running and maintaining an open-weight large language model requires suitable hardware, optimization, monitoring, and model-serving infrastructure. The organization must also evaluate output quality and factual reliability before using generated summaries for publication.

Model Comparison for Large-Scale Summarization

Feature	BERT	GPT	T5	LLaMA
Text understanding	Excellent	Excellent	Excellent	Excellent
Abstractive summarization	Limited	Excellent	Excellent	Excellent
Extractive support	Excellent	Good	Good	Good
Fine-tuning for a specific task	Good	Good	Excellent	Excellent
Instruction following	Limited	Excellent	Good	Excellent
Self-hosting	Good	Depends on	Good	Excellent

flexibility		model/deployment		
Best role in this problem	Supporting analysis	Flexible high-quality summarization	Dedicated summarization	Customizable generative summarization

Recommended Model

T5 is the most appropriate recommendation for a news organization whose primary requirement is automated summarization of thousands of articles. Its encoder-decoder sequence-to-sequence architecture directly matches the summarization task. A T5 model can be fine-tuned on a large collection of news articles and human-written summaries to produce consistent outputs.

T5 is especially appropriate when the organization wants a dedicated summarization model rather than a general conversational model. The system can be optimized for summary length, writing style, and domain vocabulary. This can support predictable processing across a large number of articles.

Practical Architecture

- Article Ingestion: collect articles from the organization’s publishing or content-management systems.
- Preprocessing: clean text, remove unnecessary elements, and prepare articles for model input.
- Content Analysis: identify article type, topic, and important information where required.
- T5 Summarization Model: generate a concise summary from each article.
- Post-processing: check length, formatting, and duplicate or irrelevant content.
- Quality and Factual Checks: verify important facts and flag uncertain summaries for review.
- Publishing Pipeline: send approved summaries to the website, application, newsletter, or internal system.

Role of Other Models

Although T5 is recommended as the main summarization model, the other models can still provide useful supporting functions. BERT can classify articles, detect relevant sentences, or support semantic matching. GPT can be used when highly flexible instructions or editorial rewriting are required. LLaMA can be considered when the organization needs open deployment, local processing, or extensive customization.

A hybrid system can therefore combine multiple models according to their strengths. However, for a dedicated and repeatable summarization workflow, T5 provides a particularly direct architecture and can be fine-tuned for the organization's specific requirements.

Conclusion

BERT is primarily an understanding model and is better suited to supporting analysis than direct abstractive summarization. GPT provides excellent flexible text generation, while LLaMA provides strong generation with open deployment and customization options. T5 is especially well matched to sequence-to-sequence summarization and can be fine-tuned for news-specific requirements. Therefore, T5 is recommended as the primary model for a large-scale automated news summarization system.